Changes to the Alaska-wide Sablefish Stock Assessment for 2026 and Summary of the Center for Independent Experts (CIE) Review of the Assessment

Daniel Goethel (Alaska Fisheries Science Center, AFSC)
and
Matt Cheng (University of Alaska-Fairbanks, UAF)

Executive Summary

An independent, external review of the Alaska-wide sablefish stock assessment was conducted by the Center for Independent Experts (CIE) in Juneau, AK from June 16 – 18, 2026. The purpose of the peer review was to determine whether the best scientific information available (BSIA) was being used to assess the Alaska sablefish population and develop robust management advice. The primary consideration for the review was how the stock assessment should be structured to handle recent NOAA Longline Survey (LLS) design changes, where the annual survey footprint was truncated beginning in 2025 (i.e., with only the Gulf of Alaska, GOA, being surveyed in odd years and the Bering Sea, BS, and Aleutian Islands, AI, being surveyed in even years). For the CIE review, a series of model building and bridging exercises were undertaken, which started with the single region continuity model (Model *25.12_1Reg_Cont*), then bridged to a new single region model (Model *26.1_1Reg*) that incorporated updates to input data and model parametrization (see Table 1 and Appendix 3B). Next, a new spatially implicit model (termed a 'fleets-as-areas', FAA, model; also referred to as an areas-as-fleets model) was developed (Model *26.2_FAA*). The parametrization of Model *26.2_FAA* matched that of the updated single region model (Model *26.1_1Reg*) but disaggregated the LLS into three regional fleets (GOA, BS, and AI) and fit regional indices and age composition data explicitly without requiring imputation for missing data (i.e., non-surveyed regions). This document briefly summarizes the six documents produced for and reviewed by the CIE panel as well as the three CIE reviewer reports (all available from the September groundfish Joint Plan Team e-Agenda).

As noted, the model development process started with the 2025 single-region continuity model (Model *25.12_1Reg_Cont*) and built towards Model *26.1_1Reg* through refined data inputs (i.e., removal of CPUE data and length compositions for data sources with age compositions as well as updated growth and weight estimates), updated model parametrizations (i.e., estimating sex-specific natural mortality and assuming logistic selectivity for the trawl fleet), and improved single region survey index creation methodology (i.e., use of the 'carryover' approach and removal of extrapolated western AI stations; see Table 1 and Appendix 3B). To address incomplete annual survey spatial coverage, various parametrizations of a spatially implicit FAA model were then explored, which built upon Model *26.1_1Reg* and added regionally disaggregated fleet structure then explored alternative selectivity options, resulting in Model *26.2_FAA* (see Appendix 3C). Although three-region and five-region spatially explicit models were also developed that estimated movement and integrated mark-recapture data, these were deemed too unstable for operational management (see Appendix 3D). To help identify robust model structures given uncertainty from the new LLS design, a spatially explicit closed-loop simulation analysis conditioned on the five-region spatial assessment was implemented to evaluate assessment performance when confronted with realistic spatial dynamics and data availability scenarios (Appendix 3E).

Both models (Model *26.1_1Reg* and Model *26.2_FAA*) displayed limited diagnostic issues, but a persistent tension remained across modeling approaches when attempting to balance rapid population growth and extreme recruitment signals from age compositions against more moderate growth observed in the LLS indices (a longstanding issue for the sablefish assessment). Similar population trends were estimated across models, with strong 2014-, 2016-, and 2017-year classes. However, Model *26.2_FAA* led to a moderate downward scaling of spawning stock biomass and a ~25% reduction in Acceptable Biological Catch (ABC) compared to Model *26.1_1Reg*. The difference in scaling highlights the potential risk of using extrapolated

survey indices in a single-region model and assuming composition data in surveyed regions are representative of the entire population, both of which may act to artificially inflate biomass and catch advice. Closed-loop simulations indicated that the spatially implicit FAA model was likely more robust and stable compared to either the single-region or fully spatial models, despite potential data limitations from sparse regional age composition data (see Appendix 3E).

The CIE reviewers endorsed the spatially implicit FAA model (Model *26.2_FAA*) as providing the BSIA for operational sablefish assessment and management advice. Moreover, the CIE reviewers noted that current assessment limitations stem from fundamental data issues, where further model complexity or alternative statistical assumptions were unlikely to lead to any meaningful improvement in model performance. The CIE panel concluded that investment in data collection, particularly increased otolith age reading and broader spatial coverage of the LLS, would provide the best avenue for improving management advice for sablefish.

Given the endorsement of the CIE reviewers and the conclusions of the assessment authors (as detailed in the SAFE document produced for the CIE review), Model *26.2_FAA* is being proposed for the basis of Alaska sablefish assessment and management advice in 2026. **For the final 2026 sablefish SAFE, results and writeups will be based on Model *26.2_FAA*, but comparisons of population trajectories (SSB, recruitment) and catch advice (7 projection scenarios) will be provided for both Model *26.2_FAA* and Model *26.1_1Reg* (Model *25.12_1Reg_Cont* will no longer be updated).**

Finally, because sablefish are assessed Alaska-wide, the resultant Alaska-wide ABC must be apportioned to management region and changes in the LLS design require updating the apportionment approach. Sablefish apportionment is based on the five-year moving average of regional biomass proportions from the LLS. Starting in 2026, the sablefish apportionment approach will use the same index carryover values as used to develop the revised single region LLS index (as in Model *26.1_1Reg*). Mainly, for non-surveyed regions in a given year, the index values are fixed (carried over) at the value from the last survey year for that region. Based on comparison of 2023 apportionment values across the old and revised method, the update in methodology has small impacts on regional apportionment (Table 8). The removal of the extrapolated western AI stations reduces the relative apportionment to the AI region, which proportionally increases apportionment to other regions. Overall, the updated apportionment approach ensures consistency with how LLS indices are constructed in the new single region model, while maintaining the long-term apportionment goals of 1) distributing catch proportional to biomass to avoid deleterious localized population impacts; and 2) avoiding large interannual fluctuations in the distribution of the ABC across management regions.

2026 Sablefish Model Development and Evaluation

A new spatially implicit fleets-as-areas (FAA) model (Model *26.2_FAA*) was introduced for the Center for Independent Experts (CIE) review in Juneau, AK from June 16 – 18, 2026 as an alternative to the updated single-region assessment framework (Model *26.1_1Reg*). Unlike the single-region approach, Model *26.2_FAA* disaggregates the NOAA longline survey (LLS) into three regional fleets—the Gulf of Alaska, Bering Sea, and Aleutian Islands—to fit regional abundance indices and age composition data explicitly. Both models were presented and reviewed by the CIE panel. The primary benefit of the spatially implicit FAA model is the ability to treat non-surveyed regions in a given year as missing data rather than relying on imputed data to generate a single aggregated index. In general, a spatially implicit FAA model can accurately represent fish populations that are reproductively well-mixed (i.e., panmictic) but that demonstrate complex movement resulting in spatially-varying availability. However, applying FAA models to populations with limited movement or complex population structure may lead to increased model bias, which must be weighed against the data demands and complexity of implementing a fully spatial assessment. Nonetheless, sablefish are likely a good candidate for a spatially implicit FAA model given their panmictic stock structure and high movement potential, despite underlying ontogenetic connectivity patterns.

The model development process involved a stepwise transition from the 2025 continuity model (Model *25.12_1Reg_Cont*) to Model *26.1_1Reg* by refining data inputs and model parametrization, including estimation of sex-specific natural mortality and improved methods for deriving the single region LLS index among other updates (see Table 1 and Appendix 3B). To address incomplete annual survey spatial coverage, various parametrizations of a spatially implicit FAA model were explored that built upon Model *26.1_1Reg,* but disaggregated fleet structure then explored alternative selectivity options, which resulted in Model *26.2_FAA* (see Appendix 3C). Although three-region and five-region spatially explicit models were also developed that estimated movement and integrated mark-recapture data, these were deemed too unstable for operational management (see Appendix 3D). To help identify robust model structures given uncertainty from the new LLS design, a spatially explicit closed-loop simulation analysis conditioned on the five-region spatial assessment was implemented to evaluate assessment performance when confronted with realistic spatial dynamics and data availability scenarios (Appendix 3E).

Evaluating the single-region model (Model *26.1_1Reg*) against the spatially implicit model (Model *26.2_FAA*) highlighted key considerations for model selection in terms of model stability, data treatment, and sample sizes. While Model *26.2_FAA* eliminates the need to impute LLS index data across non-surveyed regions, it introduces the potential for overinterpreting sparse regional composition data (see Table 2). However, the closed loop simulation indicated that the spatially implicit FAA model was likely more robust and more stable than either the single region or spatial assessment models under data conditions emulating the current survey design (Appendix 3E). When fitting the real-world data, both models exhibited a persistent tension between fitting survey indices compared to age compositions, where the former are given more emphasis via Francis reweighting and generally better fit (Table 3 describes the estimated parameters for each model and associated maximum gradient, while Table 4 shows input and effective sample sizes after Francis reweighting). Although residual patterns in the composition data remain for either model, the aggregated fits were adequate (see 2026 CIE SAFE) as were the fits to the LLS indices (Figures 1 – 3). Generally, both models have trouble reconciling the larger year classes and population growth suggested by the age composition data in recent years compared with the more tempered, but still extremely rapid, population growth observed in the LLS indices since ~2018.

Parameter estimates and population trends generally aligned across models (Table 5; Figures 4 -- 6) with both indicating strong 2014-, 2016-, and 2017-year classes, and a reduction towards average recruitment since 2020. However, Model *26.2_FAA* estimated larger 2014- and 2017-year classes and a smaller 2016-

year class (Figure 5). Alterations in recruitment estimates are due to differences in recruitment signals across the regional LLS age compositions, where the AI provides a mixture of recruitment signals (strong 2014-, 2016-, and 2017-year classes) that balances those in the BS (strong 2014 and 2016 and very strong 2017-year classes) and GOA (strong 2014 and very strong 2016-year classes). Model *26.2_FAA* results in a moderate downward scaling of spawning stock biomass (SSB) compared to Model *26.1_1Reg* (Figure 6). Ultimately, catch advice (Acceptable Biological Catch, ABC) from model *26.2_FAA* is ~25% lower than that from Model *26.1_1Reg* (Table 6). Aside from model tension resulting from trying to fit age-compositions and abundance indices simultaneously, no major diagnostic issues existed with either model (e.g., no strong retrospective patterns were observed; Figures 6 – 8).

Interestingly, an exploratory run with Model *26.2_FAA* using imputed regional 'carryover' indices (i.e., where the regional indices now summed to the aggregate index used in Model *26.1_1Reg*) led to a strong upward scaling of SSB, particularly since the start of the LLS in the mid-1990s (Appendix 3C). Thus, there may be inherent risk with using an extrapolated survey index in the assessment, which can potentially result in artificially inflating SSB and catch advice. Moreover, the phase-plane for Model *26.2_FAA* suggests that overfishing may have been occurring on the sablefish population for much of the 2000s. Therefore, the single region model may be more risk prone, if population scale is more in line with that estimated in Model *26.2_FAA*.

Extensive sensitivity runs were undertaken for both the single region (Appendix 3B) and spatially-implicit assessments (Appendix 3C), including exploration of time-varying growth and weight-at-age, continuous time-varying survey selectivity, the use of the Dirichlet-multinomial (DM) distribution for composition data, and the combination of continuous time-varying selectivity with the DM distribution. None of the alternative approaches greatly improved model performance (e.g., fits to data) and often led to model instability and degraded diagnostics (e.g., increased retrospective patterns). Thus, none were put forth for consideration as the operational assessment. Moreover, three- and five-region spatial assessments were developed and updated (Appendix 3D), illustrating interesting regional patterns in recruitment, SSB, and ontogenetic movement. However, the spatial models were not deemed insufficiently stable for operational management advice and require further exploration and testing, specifically to enable estimation of time-varying movement, regionally-varying selectivity, and how best to handle data weighting of mark-recapture data relative to the other data sources.

2026 CIE Review of the Alaska-wide Sablefish Assessment

An independent, external review of the Alaska-wide sablefish stock assessment was conducted by the Center for Independent Experts (CIE) in Juneau, AK from June 16 – 18, 2026. The purpose of the peer review was to determine whether the best scientific information available (BSIA) was being used to assess the Alaska sablefish population and develop robust management advice.

Review Purpose and Goals

The reviewers were tasked with understanding sablefish biological and fishery dynamics, appraising the assessment data inputs and limitations, and, ultimately, reviewing the stock assessment framework being utilized. The overarching goals of this review process were to:

1. Ensure the FAA stock assessment (Model *26.2_FAA*) supports the development of robust advice for use by the North Pacific Fishery Management Council based on the BSIA;
2. Provide an independent external review of this stock assessment to confirm that it meets the mandates of the Magnuson-Stevens Fisheries Conservation and Management Act (MSA) and other legal requirements;
3. Understand and identify appropriate FAA model structure and fits to key data sources (indices, landings, and age compositions);

4. Identify research needed to improve the assessment and advice for fishery managers.

Terms of Reference (ToRs) for the Peer Review

CIE reviewers are contracted to complete their independent peer review based on the ToRs. Therefore, the CIE-NOAA Fisheries review and approval process is based on whether the CIE independent reports addressed each of the ToRs.

1) Review the disseminated documents and analyses, then provide feedback on whether the fleets-as-areas (FAA) assessment provides a suitable basis for operational management advice, particularly in comparison to the current single region panmictic assessment.
   a) Provide recommendations on the most appropriate parametrization of fleet structure (number of fleets per fishery/survey, as a proxy for regions) in the FAA model.

2) Highlight potential strengths and weaknesses of the FAA assessment, including model assumptions, estimates, fits to data (survey indices and age compositions along with fishery catch-at-age data), model diagnostics, and major sources of uncertainty.
   a) Indicate whether the model provides the best scientific information available for estimating current stock status, projecting future abundance, and determining Overfishing Levels (OFLs) and Acceptable Biological Catches (ABCs).

3) Provide research recommendations for improving the existing research-oriented sablefish spatial (i.e., 3 and 5 region) assessment models, especially in the context of potential future usage in a management context.

CIE Documentation

The documents associated with the CIE review are available for download from the CIE GitHub Repo and the associated NPFMC groundfish joint plan team (JPT) e-Agenda. A new *draft* Stock Assessment and Fishery Evaluation (SAFE) report was created for the CIE review, which served as the primary document for review. The SAFE summarizes input data, model structure, and outputs for two sablefish assessment models: Model *26.1_1Reg* and Model *26.2_FAA*. The main goal of the review was to provide recommendations on which model structure provided the best scientific basis for developing operational management advice for sablefish in 2026 and beyond. The CIE review meeting was structured around comparing and contrasting performance of these two model structures.

Five appendices were developed that summarized different aspects of model building, which provided additional context for understanding the two models presented in the main SAFE document. These include a summary of the Stochastic Population Over Regional Components (SPoRC) stock assessment platform (a generalized RTMB-based R package) and model structure (Appendix 3A), single region model changes since 2025 (Appendix 3B), fleets-as-areas (FAA) model development (Appendix 3C), summary of spatial models (3D), and a summary of a spatially explicit closed loop simulation that was used to inform model structure decisions (Appendix 3E). All of the appendices were meant as sources of additional information for understanding the decisions regarding model structure in the main SAFE document.

The final documents for review by the CIE included (available here):

- 2026 Assessment of the Sablefish Stock in Alaska: 2026 Center for Independent Experts (CIE) Review Document (here)
- Appendix 3A. The Spatial Population over Regional Components (SPoRC) Model: Technical Description of the Assessment Model for Sablefish (here)

- Appendix 3B. Model Building and Bridging Exercise for the Sablefish Single Region Models (Bridging from Model *25.12* to *26.1_1Reg*) (here)
- Appendix 3C. Development of a Fleets-as-Areas (FAA) Stock Assessment Model for Alaska Sablefish (*Model 26.2_FAA*) (here)
- Appendix 3D. Results of Spatial Stock Assessment Models for Alaska Sablefish (here)
- Appendix 3E. Evaluating Alternative Assessment Structures for Alaska Sablefish Under Survey Design Changes via Closed-Loop Simulation (here)

Additionally, the presentations given during the CIE review are uploaded and available at the GitHub repository (here).

Additional sablefish background documents, which were not part of the review requirements, were also provided to reviewers and uploaded (available here). Within the background documents, the assessment documents folder was of particular value for understanding previous modeling decisions (in particular Goethel and Cheng, 2025, describing changes to the model implemented in 2025).

The data and model files (R files) used to run the assessment in SPoRC as well as model outputs are available on the GitHub repo (here). If there are any questions about models, files, or repo organization, please contact the assessment authors.

The final CIE review reports, which were developed independently by each of the three reviewers, are also available on the GitHub repo (here) and the JPT e-Agenda.

CIE Panel Main Conclusions

The CIE review of the Alaska-wide sablefish stock assessment evaluated the transition from the historical single-region panmictic model to the spatially implicit fleets-as-areas (FAA) model (specifically Model *26.2_FAA*). The reviewers were highly complimentary of the organization, transparency, and thorough preparation of the assessment team, noting that the comprehensive background documentation, well-structured presentations, and accessible GitHub repository made it easy to run and explore models independently. Moreover, the CIE Review Report submitted by Dr. Sam Subbey noted that:

> "The openness and willingness of the assessment team to engage in scientific discussion represents a model for future assessment reviews…Presentations were well prepared, discussions were constructive, and the assessment authors demonstrated a commendable willingness to critically evaluate the strengths and weaknesses of their own work…A particularly positive aspect of the review was the breadth of analyses available for evaluation."

The reviewers supported the transition from ADMB to the modernized RTMB-based SPoRC platform and agreed that the three-fleet spatially implicit fleets-as-areas (Model *26.2_FAA*) assessment represented the BSIA. Specifically, the reviewers noted that the FAA framework resulted in a pragmatic balance between parsimony and complexity. Moreover, Model *26.2_FAA* provided a robust scientific basis for operational management advice by successfully overcoming the challenges of missing regional data from the recent LLS redesign without needing to impute survey indices. Furthermore, while the spatially explicit (three- and five-region) models were recognized as promising long-term research tools with strong conceptual advantages, reviewers agreed they are currently too unstable for operational use due to data and parametrization limitations. Overall, the documentation and supporting analyses were commended for their rigor, depth, and clarity in exploring sensitivity runs and structural choices to support assessment parametrization decisions. Moreover, reviewers appreciated that many of the model uncertainties had been explored with simulation analysis (including the new spatially explicit closed loop simulation developed for the CIE review) and published in the peer-review literature. The full review reports are summarized below with specific recommendations provided in Table 7, including assessment author responses and an

outline of future plans to address each comment.

- **Operational Assessment Model (Model *26.2_FAA*):** The reviewers concluded that the spatially implicit fleets-as-areas (FAA) Model *26.2_FAA* provides the most scientifically defensible basis for operational management advice compared to the single-region Model *26.1_1Reg*. It represents the BSIA for estimating stock status, projecting future abundance, and determining Overfishing Levels (OFLs) and Acceptable Biological Catches (ABCs).
- **Handling Survey Redesign:** The primary advantage of the FAA approach is its ability to more appropriately handle the revised NOAA Longline Survey (LLS) sampling design (Gulf of Alaska in odd years, Bering Sea and Aleutian Islands in even years). Specifically, the FAA model treats missing (non-surveyed) region data for the LLS directly as missing values rather than relying on imputation or carryover values.
- **Model Structure and Fleet Parameterization:** The chosen 3-fleet configuration—splitting the LLS into Bering Sea (BS), Aleutian Islands (AI), and Gulf of Alaska (GOA) regional fleets—captures essential biological and spatial dynamics (such as ontogenetic availability) without overparameterization. Adding further fleet subdivisions or complex time-varying selectivity causes estimation instability, likely due to limited age data (see Appendix 3C, particularly Model *26.2a_FAA_Fsh+Srvy* and Model *26.2_FAA_ContTVSel_DM*).
- **Data Limitations and Structural Conflicts:** The reviewers agreed that the persistent structural conflict/tension between survey indices and age composition data was best handled through Francis reweighting, and that more complex parametrizations (full time-varying selectivity) or alternate statistical assumptions (Dirichlet multinomial) would not resolve the conflict. Ultimately, the assessment is currently limited by data quality and coverage (e.g., age sample sizes and reduced survey footprint) and not by model structure – more complexity is not the solution to data deficiencies.
- **Spatially Explicit Models:** Spatially explicit (three- and five-region) models are conceptually superior for dealing with complex sablefish ontogenetic movement patterns and spatial processes. Although the spatial models represent a promising research platform, they are currently too unstable for operational management due to overparameterization, sparse spatial age composition data, and poorly constrained movement dynamics. However, as the spatial models are further refined, they should be considered for the basis of operational management advice.

Comprehensive Summary of Recommendations

1. Assessment Model Parameterization and Diagnostics

- **Adopt Model *26.2_FAA*:** Transition to Model *26.2_FAA* for operational management advice using the current parametrization and fleet configurations, including three longline survey fleets (BS, AI, GOA) and logistic selectivity for all fleets.
- **Data Reweighting and Likelihoods:** Continue using Francis reweighting for composition data. Do not use Dirichlet-multinomial likelihoods or continuous time-varying selectivity at this time due to potential for overweighting compositional data and increased model instability.
- **Natural Mortality (*M*):** Perform sensitivity analyses on *M* to quantify the impact of its tightly constrained prior on OFL and ABC estimates. Evaluate use of alternative prior formulations (e.g., Jeffrey's non-informative priors or tagging-derived priors) via simulation.
- **Jitter Analyses:** Re-run jitter convergence tests with larger perturbations of starting values to further confirm global convergence.
- **Model Communication:** Carefully frame and explain FAA results to managers and stakeholders, focusing on fleet/availability characteristics rather than complex mathematical spatial proxies.

2. Data Streams and Survey Design

- **Survey Footprint Protection:** Maintain or expand the NOAA Longline Survey footprint. Resist further survey reductions, as diminished survey coverage will increase assessment uncertainty. Expanding coverage back to full annual surveys (and including the western Aleutian Islands) would significantly improve model robustness.
- **Discontinue Integration of Conflicting Data Streams:** Remove the trawl survey indices (due to high sampling variability and inconsistent trends) and CPUE indices (due to gear changes and lack of updated logbook data since 2022).
- **Fishery Logbooks:** Continue to analyze CPUE data when/if logbook data become available to the AFSC in the future, especially to better understand spatial catch and effort patterns. Continued development of CPUE indices provides an additional abundance index that can be used as an external (to the assessment) check on population trends and can help track spatial fleet shifts within management areas.
- **Depredation Protocols:** Formally document and standardize protocols for removing survey skates compromised by killer whale or sperm whale depredation.

3. Otolith Collection and Aging

- **Increase/Maintain Aging Capacity:** Maintain otolith collection at or above current levels (~1,200 survey and fishery otoliths aged annually, though fishery ages have recently decreased to 600) to reduce spatial and age sparseness (given that the model includes 30 ages and disaggregates the survey age compositions by region). Increase the number of otoliths aged annually (and evaluate reading archived historic otoliths). Increased otolith reading would reduce age sparseness, better inform recruitment and selectivity estimation (including the potential to implement more complex time-varying selectivity options), reduce uncertainty, and potentially enable operationalizing the spatial assessment model.
- **Staffing and Training:** Hire and train an additional otolith reader or reallocate readers from easier to age species.
- **Evaluate Sample Size Impacts:** Conduct simulation sensitivity tests with the FAA model to determine optimal regional sample sizes and evaluate how reduced age sample sizes impact model stability.
- **Aging Error Research:** Further investigate sources of uncertainty in otolith aging. Evaluate whether reducing assumed aging error improves model stability.

4. Research on Spatially Explicit Models and Biology

- **Observer Data Analysis:** Examine observer age compositions spatially to determine if spatial shifts in fishing effort caused historical "plus group" (age 31+) anomalies.
- **Tagging Programs & Otolith Recovery:** Expand mark-recapture programs (including pop-up satellite tags and juvenile tagging) to better estimate age-specific movement rates and regional connectivity. Increase incentives for fishers/processors (e.g., higher lottery payouts or more tickets) to return tagged fish with otoliths.
- **Calcein Injection:** Consider injecting tagged survey fish with calcein to assist in verifying annual growth annuli.
- **Environmental and Dispersal Modeling:** Expand hydrodynamic/larval dispersal models (e.g., MOM-6) to include the Bering Sea and Aleutian Islands to better understand spatial recruitment connectivity.
- **Spatial Reference Points:** Investigate and resolve the mismatch between spatial model outputs and management requirements, including how to define Tier 3 reference points across multiple spatial areas.

5. Software Engineering (SPoRC Package)

- **Technical Documentation:** Develop and maintain a comprehensive technical design document for the SPoRC package to assist future developers, succession planning, and collaboration.
- **Unit Testing:** Expand automated formal unit tests within the SPoRC codebase and run them routinely (e.g., automated weekly builds).

Plans for the 2026 Alaska-wide Sablefish Stock Assessment

Given the endorsement of the CIE reviewers and the conclusions of the assessment authors (as detailed in the SAFE document produced for the CIE review), Model *26.2_FAA* is being proposed for the basis of Alaska sablefish assessment and management advice in 2026. Based on closed loop simulations and model performance when fit to the real-world data, Model *26.2_FAA* is deemed to provide a more robust model that better handles extant data limitations associated with the LLS redesign. Although model scale of SSB and resultant catch advice have declined moderately compared to the single region Model *26.1_1Reg*, the change in scale is not surprising given that Model *26.2_FAA* no longer includes extrapolated indices which appear to artificially inflate model scale. **For the final 2026 sablefish SAFE, results and writeups will be based on Model *26.2_FAA*, but comparisons of population trajectories (SSB, recruitment) and catch advice (7 projection scenarios) will be provided for both Model *26.2_FAA* and Model *26.1_1Reg* (Model *25.12_1Reg_Cont* will no longer be updated).**

Response to SSC and JPT Comments

The following summarizes feedback and planned analyses to address recommendations from the groundfish Joint Plant Team (JPT) and the Science and Statistical Committee (SSC) since the last operational stock assessment was reviewed (2024). Review body comments are in italics with author responses provided directly after each comment.

NPFMC SSC Comments (December 2024)

The SSC agrees with the author and JGPT that exploring additional sex-specific dynamics, in particular disaggregating age data by sex, should be a priority. This may help address troublesome patterns in (sex-aggregated) age composition residuals and (sex-disaggregated) length composition residuals, which may reflect some combination of time-varying (e.g. density-dependent) growth, time-varying selectivity, differences in M between sexes, deviations from a 50:50 sex ratio, and aging errors.

The 2025 model updates as integrated into Model *25.12_1Reg_Cont* aimed to better model sex-specific dynamics, primarily through fitting sex-disaggregated composition data. Moreover, all models presented in 2026 fit sex-disaggregated compositions and estimate sex-specific natural mortality.

The SSC further suggests explorations of models that allow for variations in growth over time, given the results from Cheng et al. (2024) that provide strong evidence of density-dependent growth. Variability in growth could result in the residual patterns noted above.

Sensitivity models were presented in both 2025 and 2026 that explore time-varying growth and weight-at-age as estimated in Cheng et al. (2024). Although these models provide interesting insight and have important impacts on SSB, they do not improve model performance or reduce residual patterns. Moreover, there is considerable uncertainty in the time-varying growth estimates, given sample size limitations. Thus, these models are not put forward as the basis of management advice.

The SSC supports the plan team recommendation to explore the use of time-varying selectivity in the model to address the poor fit to the fishery CPUE index and fishery catch-at-age data.

Multiple approaches to time-varying selectivity were presented as sensitivity runs in 2025 and 2026. Although representing a promising avenue of research, continuous time-varying selectivity formulations do not greatly improve fits to the age compositions and often reduce fits to the longline survey indices. Moreover, they tend to cause model instability, given the number of model partitions included for sablefish (i.e., sex, age/length, fleet, year). The CIE reviewers recommended maintaining the status quo for selectivity parametrization, which includes two time blocks for the fishery and survey. Moreover, the CIE reviewers explicitly stated that continuous time-varying selectivity should not be utilized at this time. Thus, the time blocked selectivity approach has been maintained for all 2026 models.

The SSC re-iterates a recommendation from last year to explore the Hulson and Williams (2024) bootstrapping approach for determining appropriate input sample sizes for compositional data.

The Hulson and Williams (2024) ISS approach has not been fully developed for longline survey data nor has it been developed for fishery compositions. Until these methods are resolved for all data sources, it was not deemed appropriate to implement as it may introduce model issues (i.e., when only one type of composition data uses a bootstrapped ISS but others do not). To address these concerns, a new time-varying ISS approach was introduced that scaled ISS relative to the number of samples in a given year and fleet (see Appendix 3B). However, regardless of approach for setting ISS, the ISS likely has limited impact on model outputs or performance given the degree of rescaling from ISS to ESS that occurs with the Francis reweighting.

The SSC appreciates the clarification of how the recruitment bias correction is currently handled in the model but encourages further explorations regarding the treatment of recruitment variances. As noted last year: "The SSC encourages the use of an appropriate sigma constraining recruitment but notes that the maximum likelihood estimate of a random effects variance is negatively biased. This can be avoided by iteratively tuning in a maximum likelihood framework (per the approach of Methot and Taylor) or by using a full Bayesian analysis such that the recruitment deviations are integrated out."

The iterative tuning method of Methot and Taylor was integrated and used to determine the recruitment bias correction parameters, but had little impact on values. Work is ongoing to move towards treatment of recruitment as random effects, given that the RTMB SPoRC model is capable of handling REs. However, initial explorations indicate that integration of REs greatly increases model run time. This work will continue to determine if recruitment REs are a feasible alternative in the future.

The SSC shares PT concerns over the loss of logbook data and supports the PT recommendation to consider developing a standardized fishery CPUE index using observer data only, especially if the longline survey cannot be conducted each year.

Investigations of the observer data suggest that the sample sizes are likely not sufficient to develop a robust CPUE index and are often extremely limited in certain regions (e.g., the AI in 2025 had no observed sablefish targeting trips).

The SSC encourages further work on skip spawning to assess its potential impact on reproductive potential, perhaps through the development of an index for the ESP.

Until resources are available to collect sablefish maturity estimates during spawning periods (winter), assessment of skipped spawning is not feasible.

The SSC encourages assessment authors to seek out local knowledge regarding fishery CPUE, in particular regarding any issues with the performance of slinky pots that may affect CPUE.

The assessment authors hold annual meetings with three industry groups and use these opportunities to discuss changes in the fishery and on the water observations. We are not currently aware of any issues with slinky pots that may affect CPUE, but will continue to discuss fishing practices with stakeholders.

Joint Groundfish Plan Team Comments (November 2024)

The Teams recommended the author explore the potential impact of time-varying selectivity, either by directly modeling it as a time-varying process or by mitigating its impact on other parameters, such as by exploring changes in the set of ages over which the age-length comps are fitted. The Teams recommended exploration of data-weighting methods that can be estimated jointly with changes in the variance of the time-varying selectivity parameters.

See response to SSC comment above regarding time-varying selectivity. For 2026, continuous selectivity parametrizations were also combined with the self-weighting Dirichlet-Multinomial likelihood for composition data. The resulting model tended to overfit the composition data at the expense of the LLS index data, resulting in extremely volatile recruitment estimates. The CIE reviewers noted explicitly that time-varying selectivity and the DM likelihood should not be utilized for sablefish at this time. As such, the models put forward for 2026 do not utilize either methodology.

The Teams recommended that the author perform a runs test of randomness to test autocorrelation in the fits to the indices.

We now output and report an index runs test as a residual diagnostic for all model runs.

If there are sufficient data, the Teams recommended exploring ways to update the standardized fishery CPUE index using only observer data.

See response to similar SSC comment above. Observer data is insufficient to produce a reliable CPUE index that adequately represents trends across all management regions.

NPFMC SSC Comments (October 2025)

The SSC also recommends including biennial survey impacts on extrapolation in the 2026 CIE review.

The LLS design changes were a primary focus of the CIE review and impetus for moving to the new FAA model.

Document Francis reweighting and effective sample sizes across datasets.

A table is now included for all model runs that provides the SE for indices and the mean ISS and ESS (after Francis reweighting) for compositional data sets.

Compare empirical length and weight-at-age directly with model estimates.

Comparison of the external estimates of growth and WAA with empirical values has not been undertaken, but is a topic that will be explored in the future. However, empirical WAA values are likely to be noisy given the relatively low level of weight samples (compared to length samples) taken on the LLS. Model sensitivity runs with time-varying growth and weight-at-age using the methods of Cheng et al. (2025) are presented in Appendices 3B and 3C.

Continue evaluation of flexible fishery selectivity and alternative composition likelihoods.

See responses to similar comments above.

Continue development of spatial models and sex-specific mortality.

The spatial models were updated for 2026 and sex-specific natural mortality is estimated in all new models put forward in 2026.

Continue exploring approaches for using fishery composition data to inform removals only rather than population structure or recruitment.

It is unclear how this would be accomplished as composition data inherently reflects the age structure of the population along with recruitment, as modified through availability and gear selectivity. Further guidance is requested on how this might be achieved.

Explore inclusion of environmental covariates in R-TMB/SPoRC.

Although environmental covariates are possible to include in SPoRC, there are no reliable covariates that have been identified for sablefish.

Model proportions of age-groups by area using spatial approaches.

The spatial assessment explicitly models compositions by region.

Clarifying the likelihood for the prior on M on the scale of M, not logM, and the other likelihood component.

It is unclear exactly what is being requested with this comment.

Avoid using an ad hoc approach to down weighting anomalous size comp data points.

The only downweighting used is Francis reweighting, which is a relatively objective and well-established method for data weighting. It is unclear what this recommendation is suggesting.

Consider the reduced survey frequency in the risk tables

For 2026, survey frequency and footprint will be specifically addressed in the risk table.

Joint Plan Team Comments (September 2025)

The Teams recommended clarifying and documenting the results of Francis reweighting and include tables of input sample sizes and final effective sample sizes. This also extends to the indices since multipliers appear to remain in use (i.e., "lambdas"; it is better to report the variance or CVs implied).

As noted in the SSC comments above, these tables are now provided for all model runs.

The Teams recommended comparing empirical length and weight-at-age directly with modeled estimates.

As noted in SSC comments above, this is a research priority but has not yet been addressed.

The Teams recommended testing approaches where fishery composition data inform removals only rather than population structure or recruitment.

As noted in the SSC comments above, it is unclear exactly how this would be undertaken and further guidance is requested.

Apportionment

Because sablefish are assessed Alaska-wide but managed across 6 management units spanning three fishery management plans (FMPs; GOA, BS, and AI), the resultant Alaska-wide ABC must be apportioned to management region. Currently, sablefish apportionment is based on the five-year moving average of regional biomass proportions from the LLS. Because the LLS is not synoptic, imputation is required to fill in index values for non-surveyed regions in a given year to enable calculation of annual regional apportionment. Although apportionment does not require developing a single region-wide 'index', it does require index values for each region. Survey index imputation is necessary for apportionment even though the recommended spatially-implicit assessment (Model *26.2_FAA*) was specifically designed to avoid the need to use imputed survey index values.

Previously, the survey apportionment method utilized the same assumptions as were used to construct the single region LLS index (as used in Model *25.12_1Reg_Cont*). Mainly, for regions that were not surveyed in a given year (i.e., BS or AI), associated index values in off years were constructed by extrapolating from the previous value in that region based on the year over year growth in the GOA. However, with the new LLS design starting in 2025, this approach is no longer feasible because the GOA is no longer surveyed annually.

Thus, starting in 2026, the sablefish apportionment approach will use the same index carryover values as used to develop the revised single region LLS index (as in Model *26.1_1Reg*). As documented in Appendix 3A, the new index construction maintains a carryover value for non-surveyed regions (i.e., if a region is not surveyed, the index value for that region is held constant at the value from that region the last time it was surveyed). Moreover, for the construction of the AI index, previously extrapolated western AI stations -- which have not been surveyed since the early 1990s -- will be removed. Thus, the regional survey biomass (i.e., relative population weight, RPW) values used in the five-year average to determine apportionment are produced using the same methods as the associated regional survey numbers (i.e., relative population numbers, RPN) that are summed to construct the single region index used in Model *26.1_1Reg*. Thus, the apportionment relies on carryover values, as this is the most reliable way to develop a synoptic evaluation of regional biomass proportions from a spatially incomplete annual survey.

To provide a comparison of how the new survey index construction impacts apportionment across regions, we calculated apportionment using the previous approach (using GOA growth to extrapolate and including western AI extrapolated stations) and the new carryover approach (using carryover regional values and dropping the western AI extrapolated stations) since 2016, with emphasis on 2023 as the most recent year that a full comparison was possible. Because there was no survey in 2024 and 2025 was the first year of the new survey design, using 2023 as an exemplar year provided the most direct comparison of how the two methods might differ. For both apportionment calculations, the five-year average apportionment included a moving average of the five previous years (e.g., for 2023 apportionment, the 2019—2023 LLS index value were used in the calculation).

Results demonstrate that the change in index construction method has small impacts on regional apportionment (Figure 1; Table 8). The removal of the extrapolated western AI stations reduces the relative contribution of the AI, which proportionally increases apportionment to all other regions for the new proposed method. Of note, the BS proportions are scaled upwards due to the change in the AI but scaled downwards by a similar amount due to the switch to the carryover values. Because the carryover approach leads to a more moderate recent rate of increase in the BS (compared to extrapolating based on GOA year

over year change), the BS index is generally lower. Ultimately, the two influences on the BS proportion act to essentially cancel one another out leading to little change in the BS apportionment, while the apportionment values in the GOA increase moderately across methods. Overall, the updated apportionment approach ensures consistency with how LLS indices are constructed in the new single region model, while maintaining the long-term apportionment goals of 1) distributing catch proportional to biomass to avoid deleterious localized population impacts; and 2) avoiding large interannual fluctuations in the distribution of the ABC across management regions. Values for 2025 apportionment under the proposed approach are similar to the 2023 values with minor changes due to updated GOA survey data for 2025 (Table 8), while similar moderate changes are expected for 2026 with the addition of BS and AI survey data.

Tables

Table 1. Summary of the single region model bridging and model changes from Model *25.12_1Reg_Cont* to Model *26.1_1Reg*.

Model Group	*Model Name*	*Major Changes*	*Conclusions*
Continuity Model	*25.12_1Reg_Cont*	(*25.12*) None.	The continuity model retains minor instability due to selectivity parametrizations and conflicting signals among age and length data as well as the various indices; further model updates are warranted to improve performance and increase stability.
Model Parametrization (Section 1)	*26.1a_1Reg_Cont_Upd*	(*25.12*+) Remove selectivity links, update selectivity priors, and reduce survey CV.	Improved selectivity parametrization greatly improved model stability, while decreased survey CV led to minor improvements in fit to LLS index; moderate rescaling resulted from an increase in the *M* estimate that led to increases in SSB and catch advice.
	26.1b_1Reg _ISS	(*26.1a*+) Implement relative scaled input sample size (ISS) across years.	The scaled ISS had minimal impact on estimates (e.g., selectivity) but led to minor improvements in fits to composition data and mean age.
	26.1c_1Reg_TF_log	(*26.1b*+) Implement logistic selectivity for the trawl fishery.	Implementing the logistic function for trawl fishery selectivity improved model stability and led to minimal improvements to trawl fishery length composition fits, particularly larger females.
Biology (Section 2)	*26.1d_1Reg_Upd_Bio*	(*26.1c*+) Utilize updated length and weight parameters.	Led to minor scaling changes in biomass due to slight decrease in mean weight of fish under 15 years of age.
	26.1e_1Reg_Upd_Sex_M	(*26.1d*+) Estimate sex-specific natural mortality (*M*).	Caused moderate reduction in *M* estimates across ages, which led to increases in *F* along with reductions in SSB and catch advice.
Data Usage (Section 3)	*26.1f_1Reg_No_CPUE*	(*26.1e*+) Remove the fishery CPUE index.	Resulted in minor improvements in fit to the LLS index and changes to estimates of fishery selectivity (particularly in the pre-1995 block).
	26.1g_1Reg_No_Len_2_Blks_F	(*26.1f*+) Drop fixed gear fishery and NOAA LLS lengths along with associated early fishery time block.	Improved model stability considerably and led to minor changes to fishery selectivity and male *M*, leading to increases in *F* and downward scaling of SSB and catch advice; fits to early age compositions improved slightly.
	26.1h_1Reg_LLS_Carry	(*26.1g*+) Implement 'carryover' approach for NOAA LLS calculation in all years.	The 'carryover' LLS index construction approach leads to a slightly different index trajectory that is more steadily increasing in recent years, while the model is better able to fit this trend; the lower initial index values and steady recent increases lead to an upward scaling of SSB and catch advice.
	26.1i_1Reg_No_wAI_LLS	(*26.1h*+) Drop extrapolated western AI stations from the NOAA LLS index.	Removal of the extrapolated western AI stations causes the LLS index to be scaled downwards and tampers the rate of increase in recent years (given that the increasing catch rates in the AI cause the extrapolation to have a stronger impact), which results in a minor reduction in recent SSB and catch advice.
Final 1 Region Model	***26.1_1Reg***	(*26.1i*) None.	Improved parametrization and removal of uncertain and potentially contradictory data sets greatly improved model stability; updates to the LLS index construction improve consistency and better reflect observed dynamics; and improved biological inputs and parametrization likely help to better reflect sex-specific sablefish dynamics.

Table 2. Summary of tradeoffs between using a single region (1Reg), spatially implicit (Fleets-as-Areas, FAA), and muti-region (spatial) models for sablefish.

Model	*Benefits*	*Issues*	*MSE Conclusions*	*Tradeoffs*
Single Region Model (1Reg)	Higher sample sizes. Low model complexity. Intuitive and transparent results. Limited change from status quo assessment methodology.	Requires fitting extrapolated data for the Longline Survey Index. Implicit over-weighting of GOA composition data during aggregation or high interannual variability due to reduced survey footprint. Little signal from BS/AI compositions, because GOA given higher weight (RPNs or catch) in data aggregation. Ignores spatial structure.	Moderate SSB bias and moderate instability due to the survey extrapolation and aggregated composition data but impacts on resulting management advice may be negligible.	Higher sample sizes but requires fitting the extrapolated longline survey index and age compositions primarily represent the recruitment signals from the GOA due to data aggregation assumptions.
Spatially Implicit Model (Fleets-as-Areas, FAA)	Fitting non-extrapolated Longline Survey index. Fitting disaggregated survey compositions. Better accounting for signals from BS/AI compositions on recent year class strength. Implicitly accounting for ontogenetic movement dynamics	Low sample sizes and sparse data are likely for BS/AI age compositions. Increased instability may exist due to added complexity (estimating extra q and selectivity parameters for 'regional' survey fleets). Non-intuitive estimates may result that are difficult to interpret (e.g., due to using multiple fleets that are interacting with a single underlying population – not multiple populations/regions). Selectivity may need to be more flexible to address ontogenetic dynamics and changes in availability across age classes.	Low SSB bias and high model stability with improved performance over other structures when survey spatial coverage decreases, but adequate flexibility in selectivity parametrizations is likely needed to ensure good performance.	Able to fit the regional longline survey indices without aggregation and likely provide more robust outputs when survey coverage decreases, but sample size limitations may prevent adequate selectivity flexibility to address ontogenetic availability issues due to model instability.
Spatial Model (3 or 5 Region)	Fit data at the regional scale it was collected. Better account for signals from each region. More intuitive than FAA. Explicitly account for ontogenetic movement dynamics.	Increased sample size limitations. Increased parametrization decisions related to spatial processes. Lack of best practices for fitting tagging data (e.g., dealing with tag mixing issues and data weighting). Increased parameter correlation, particularly between regional recruitment parameters and movement. Diagnostics can be difficult to interpret.	High instability due to overparameterization and low sample sizes but best represents localized dynamics.	Best represents spatial dynamics but suffers from high model instability due to data sparsity that requires extremely careful parametrization, while appropriate handling of tagging data is difficult and potentially subjective.

		Increased instability.		

Table 3. Summary of the parameters estimated within each assessment model and the associated maximum gradient component.

Parameter Name	***25.12_1Reg_Cont***	***26.1_1Reg***	***26.2_FAA***
Catchability	5	2	4
Mean recruitment	1	1	1
Natural Mortality	1	2	2
Initial Deviations	28	28	28
Recruitment Deviations	65	65	65
Mean Fishing Mortality	2	2	2
Fishing Mortality Deviations	129	129	129
Fishery Selectivity	14	12	12
Survey Selectivity	11	12	28
Total	256	253	271
Maximum Gradient	2.64 e -05	2.91 e -10	7.64 e -10

Table 4. The assumed error terms for each data source used in the assessment models. Fit to catch and survey data assumed lognormal error with an associated assumed standard error (SE). Fit to composition data (ages and lengths) assumed a multinomial distribution with associated sample sizes (input sample size, ISS, and final 'effective' sample size, ESS). Each model used Francis data reweighting to 'right-weight' the composition data relative to the indices (Francis, 2011), which resulted in a weighting factor (λ , Data Wt) used to scale the ISS to achieve the final ESS. Values represent the mean across all years of associated data, where the yearly SE can be found in Table 3.6 of the CIE SAFE document and the yearly ISS can be found in Table 3.5 (associated yearly ESS can be calculated by multiplying the ISS by the respective Data_Wt for the given data source).

Data Source	*25.12_1Reg_Cont*				*26.1_1Reg*				*26.2_FAA*			
	SE	**ISS**	**Data Wt**	**ESS**	**SE**	**ISS**	**Data Wt**	**ESS**	**SE**	**ISS**	**Data Wt**	**ESS**
Fixed Gear Catch	0.05	--	--	--	0.05	--	--	--	0.05	--	--	--
Trawl Catch	0.05	--	--	--	0.05	--	--	--	0.05	--	--	--
Fixed Gear Fishery CPUE	0.20	--	--	--	--	--	--	--	--	--	--	--
NOAA Longline Survey RPN BS	--	--	--	--	--	--	--	--	0.20	--	--	--
NOAA Longline Survey RPN AI	--	--	--	--	--	--	--	--	0.24	--	--	--
NOAA Longline Survey RPN GOA	--	--	--	--	--	--	--	--	0.08	--	--	--
NOAA Longline Survey RPN AK	0.09	--	--	--	0.07	--	--	--	--	--	--	--
Coop Survey RPN	0.09	--	--	--	0.07	--	--	--	0.07	--	--	--
Fixed Gear Age Composition	--	20	0.80	16	--	145	0.12	18	--	145	0.04	6
Fixed Gear Size Composition	--	20	3.48	70								
NOAA Longline Survey Age Composition BS	--	--	--	--	--	--	--	--	--	37	0.51	19
NOAA Longline Survey Age Composition AI	--	--	--	--	--	--	--	--	--	39	1.91	75
NOAA Longline Survey Age Composition GOA	--	--	--	--	--	--	--	--	--	77	0.29	23
NOAA Longline Survey Age Composition AK	--	20	3.54	71	--	171	0.28	47	--	--	--	--
NOAA Longline Survey Size Composition AK	--	20	2.29	46	--	--	--	--	--	--	--	--
Japanese Longline Survey Age Composition	--	20	0.64	13	--	83	0.17	14	--	83	0.18	15
Japanese Longline Survey Size Composition	--	20	1.03	21	--	55	0.34	19	--	55	0.33	18
Trawl Fishery Size Composition	--	20	0.35	7	--	45	0.25	12	--	45	0.14	7

Table 5. Key scaling parameter estimates along with their uncertainty, including 95% credible intervals from MCMC analysis for each model. Recruitment is in millions of fish. All parameter estimates are in ln-space. For Model 25.12_1Reg, only a single natural mortality (*M*) is estimated and shared across sexes.

	25.12_1Reg_Cont		***26.1_1Reg***		***26.2_FAA***	
Parameter	**Mean (MLE)**	**SD (MLE)**	**Mean (MLE)**	**SD (MLE)**	**Mean (MLE)**	**SD (MLE)**
ln_Mean_Recruit	2.95	0.17	3.03	0.13	2.82	0.11
ln_M_Female	-2.38	0.09	-2.39	0.07	-2.43	0.07
ln_M_Male	--	--	-2.41	0.07	-2.44	0.08
ln_q NOAA Longline Survey	2.06	0.12	1.90	0.10	--	--
ln_q NOAA Longline Survey BS	--	--	--	--	0.55	0.10
ln_q NOAA Longline Survey AI	--	--	--	--	0.33	0.10
ln_q NOAA Longline Survey GOA	--	--	--	--	2.20	0.08
ln_q Japanese Longline Survey	1.68	0.11	1.71	0.09	2.08	0.08

Table 6. Summary of ABC and OFL for each assessment model compared to those output during the last operational assessment (2024). The values were provided as part of the *draft* 2026 CIE SAFE report and are not meant for use as management advice.

Quantity/Status	Specified *last* SAFE for Model *23.5*:		Estimated *this* SAFE for Model *26.1_1Reg*:		Estimated *this* SAFE for Model *26.2_FAA*:	
	2025	2026*	2026	2027**	2026	2027^
M Female (estimated)[#]	0.11		0.092		0.088	
M Male (estimated)	--		0.090		0.087	
Average Recruitment (millions of fish)	26		21		18	
Tier	3a		3a		3a	
Projected Total Biomass (Age 2+; kt)	705	696	628	611	469	453
Projected Female Spawning Biomass (kt)	220	241	237	253	205	211
$B_{100\%}$ (kt)	303	303	348	348	306	306
$B_{40\%}$ (kt)	121	121	139	139	122	122
$B_{35\%}$ (kt)	106	106	122	122	107	107
F_{OFL}	0.10	0.10	0.09	0.09	0.10	0.10
$maxF_{ABC}$	0.09	0.09	0.08	0.08	0.08	0.08
F_{ABC}	0.09	0.09	0.08	0.08	0.08	0.08
OFL (kt)	59	58	48	46	38	35
max ABC (kt)	50	50	41	40	32	30
ABC (kt)	50	50	41	40	32	30
Status	As determined *last* year for:		As determined *this* year for:		As determined *this* year for:	
	2023	2024	2024	2025	2024	2025
Overfishing	No	n/a	No	n/a	No	n/a
Overfished	n/a	No	n/a	No	n/a	No
Approaching overfished	n/a	No	n/a	No	n/a	No

*The 2024 SAFE projections of future SSB and biomass were based on specified catch of 25,000 mt in 2025 (a yield ratio of 0.49 was assumed based on a 2024 extrapolated catch of 23,000 mt and an ABC of 47,000 mt). Specified catch projections are utilized in response to management requests for a more accurate two-year projection. SSB and biomass are less than presented when the maximum ABC is removed in 2025.

**The 2026 CIE SAFE Model *26.1_1Reg* projections were based on specified catch of 20,000 mt in 2026 (a yield ratio of 0.49 was assumed based on a 2025 catch of 23,200 mt and an ABC of 47,600 mt).

^The 2026 CIE SAFE Model *26.2_FAA* projections were based on specified catch of 16,000 mt in 2026 (a yield ratio of 0.49 was assumed based on a 2025 catch of 23,200 mt and an ABC of 47,600 mt).

[#]A single female natural mortality parameter is estimated in Model *23.5* and is used for both sexes.

Table 7. Summary of CIE reviewer recommendations for improving the Alaksa sablefish stock assessment model.

Category	Primary Target / Issue	Specific Recommendation	Objective / Expected Benefit	Author Response	Planned Action
Model Structure	Operational Model Choice	Adopt Model *26.2_FAA* with 3 survey fleets (BS, AI, GOA) and logistic selectivity.	Provides BSIA; handles alternating biennial survey coverage without data extrapolation.	The authors agree and plan to put forth the new Model *26.2_FAA* for sablefish.	The authors are proposing Model *26.2_FAA* for 2026 advice.
	Natural Mortality (*M*)	Conduct sensitivity analyses on *M* values and evaluate alternative prior formulations (e.g., Jeffrey's priors, this should be done via simulation analysis before testing operationally).	Quantifies impact of *M* prior on OFL/ABC and provides better understanding of weak parameter identifiability.	An *M* sensitivity analysis was performed as part of the 2025 updates and demonstrated the expected result – *M* is a critical model scalar that directly scales SSB and ABC. The work in 2025 extensively explored appropriate *M* priors and we believe the current parametrization is adequate, given the extensive uncertainty and associated model instability in estimating *M*.	Integrating alternative prior formulations into SPoRC will be explored over the next year. When available, the authors will explore *M* sensitivity again and identify whether new prior formulations are warranted (using simulation analysis).
	Selectivity in the FAA Model	Use of dome shaped selectivity should continue to be investigated as more age composition data for the BS and AI become available in the coming years. Similarly, there should be further investigation into different time blocks for the FAA assessment. The use of sex-specific spline selectivity could be investigated.	Selectivity curves were limited to logistic. This is an appropriate restriction, however it does limit the capacity of the FAA model to capture spatial aspects (such as the heightened presence of juveniles and the old plus class fish in the Aleutians). Various forms of domed selectivity would allow the model to have further flexibility to fit the age compositions but also add instability. Alternate time blocks would help to demonstrate if there are other formulations that result in stable parameter estimation.	A number of domed selectivity FAA model sensitivity runs were presented at the CIE, including gamma and semi-parametric continuous time selectivity. Although these models helped better account for availability in the BS and AI, the models were increasingly unstable (i.e., demonstrated degraded retrospective performance) and did not significantly improve fits to the age composition data. Similarly, various time blocks were explored for the survey and fishery as well as continuous selectivity options. None of these model alternatives showed significantly improved diagnostics and were often more unstable.	The authors will continue to explore models with domed selectivity and alternate time blocks, including continuous time and semi-parametric variants. However, many of these models add instability and undue complexity and will not be put forth for management advice unless model performance improves significantly.
	Recruitment	Explore episodic recruitment in the model structure.	The interannual changes in survey age composition data (GOA odd years,	The authors have discussed integration of recruitment stanzas	Sensitivity runs with recruitment stanzas will be explored in the

	Parametrization		BS/AI even years) likely causes underestimation of recruitment in high recruitment years and overestimation in low recruitment years (shrinkage towards mean recruitment in recent years). Including recruitment stanzas (time-varying R_0) might alleviate the over/under-estimation issue to some degree, but the information to parameterize this could be limited.	to reflect high or low recruitment periods. However, identification of the associated temporal break points may be subjective, and, as noted, adding additional recruitment parameters is likely to add model instability. Moreover, the introduction of recruitment stanzas has important implications for management biomass reference points, which are based on an assumed time period of average recruitment. Adding stanzas would require a decision as to which stanza to use for calculation of reference points.	coming year but are unlikely to be stable enough for use in management. Moreover, the recruitment variance term (σ_R = 0.9) was chosen to allow extreme fluctuations in recruitment deviations and should be flexible enough to allow recruitment to vary widely and implicitly account for the cyclical changes in sablefish recruitment.
Diagnostics	Jitter Analyses	Re-run jitter convergence tests with larger perturbations of starting values.	Larger perturbations will help further confirm global convergence.	Initial jitter runs were set up following general AFSC guidelines but can easily be altered to better test convergence.	Future jitter diagnostics will utilize larger perturbations, as recommended.
Data Inputs	LLS RPNs vs. RPWs	Further explorations of whether it is best to fit the LLS index in numbers or weight would be useful.	The LLS is currently fit as a numbers-based index (RPNs). In years of strong recruitment, the RPN can be dominated by small fish. It might be preferable to fit the index in weight (RPW) to better reflect population biomass.	The 2016 CIE review recommended fitting the index as RPNs to better reflect recruitment signals (at a time when recruitment had been low for over a decade). At the time, the assessment team evaluated fitting RPNs vs. RPWs and concluded that converting numbers to weight would add uncertainty to the assessment outputs (given the low number of fish weight samples). Moreover, the age composition data already provides information on the proportions of younger and older fish.	The authors do not currently plan to reevaluate this model assumption, given that a thorough analysis was undertaken during the last CIE review. As noted, RPWs are much more uncertain due to weight sampling protocols and there is no reason to believe that fitting RPWs would provide a significantly different interpretation of population trajectory or alter model performance.
	Fishery CPUE & Trawl Survey	Discontinue use of trawl survey indices and commercial CPUE data.	Improves retrospective pattern diagnostics and parsimony by removing conflicting/unreliable data.	The authors agree with this approach and neither index has been included in either of the proposed models.	Moving forward both indices will be monitored in the SAFE, but neither will be included in the assessment.

Survey Design	Depredation Protocols	Formally document and standardize protocols for removing survey skates affected by whale depredation.	Ensures consistency, transparency, and reproducibility in survey index calculations.	The LLS team is currently working on more objective ways to account for whale depredation and agree that more formal documentation will be critical once methods are finalized.	Important changes to the LLS index construction and protocols are being explored and once finalized, will be documented in a NOAA Technical Memorandum.
	Longline Survey Footprint	Maintain survey footprint and prevent further spatial/station reductions; evaluate expanding back to annual surveys or including WAI.	Reduces biomass trend lag, stabilizes recruitment estimates, and lowers assessment uncertainty.	The authors and LLS team fully agree with this recommendation, and no plans are currently under consideration to change the survey footprint from the 2025 and 2026 levels. Further expansion would likely necessitate further dedicated funding from NOAA and is not under consideration at this time.	Expansion of the LLS would be considered in the future pending long-term funding acquisition.
Data Collection	Otolith Aging Capacity	Increase the number of otoliths aged annually (utilizing archived otoliths) and hire/train an additional reader.	Improves age composition sample sizes across all regions and sexes, aiding model fit.	Although the authors support this recommendation, the AFSC Age and Growth Program (AGP) is currently working at full capacity and is unlikely to be able to read more sablefish otoliths.	If AGP capacity increases in the future, the authors will work with age readers to identify appropriate increases in target sablefish otoliths to read.
	Tagging & Otolith Recovery	Enhance tag return incentives (e.g., larger lottery prizes) for returning tagged fish with otoliths.	Increases known-age tag recoveries to refine aging error matrices and movement estimates.	Unfortunately, funding to increase tag return incentives is not currently available. Given the pace of work on commercial vessels and processing plants, the likelihood of attaining otoliths along with tags is likely low, though this would be a valuable source of information. We do currently attain approximately 12% of tag returns from the LLS, where we collect the otoliths for future ageing. However, as noted above, the capacity to age more otoliths by the AGP does not currently exist.	Options will be explored for obtaining otoliths from recaptured fish, pending improved capacity to age otoliths by the AGP.
	Age Validation	Inject a subset of tagged survey fish with calcein and explore trends in ageing error with region/depth.	Assists in verifying annual growth annuli and reduces otolith aging error.	This would require extensive collaboration with the LLS and AGP teams to develop a special project that may not be feasible	The survey team and AGP will discuss the potential cost/benefits of this recommendation to determine if it would provide

				given the current LLS vessel limitations.	significant improvement to current ageing protocols and/or would be feasible on the LLS. The toxicity of calcein would need to be considered for sablefish being sent to market.
	Age Sample Size Impacts	Perform simulation sensitivity analysis to identify cost/benefit of increased age reading on assessment performance.	Will determine the expected benefit from increasing aging capacity and where otolith collection might have biggest impact spatially.	In order to fully test the value of aging data, a high spatial resolution operating model (OM) is necessary to account for the spatiotemporal nature (sparsity and correlation) of age samples, especially given the spatial domain of the assessment and low level of sampling in certain regions (e.g., the AI had no otoliths taken by observers in 2025). A simple simulation based on effective sample size (ESS) by region (i.e., increasing or decreasing multinomial ESS for simulating age compositions as is typically done) does not provide a robust conclusion as to the value of additional aging data. Indeed, the results are more or less pre-determined where higher ESS (more samples) will inherently reduce assessment uncertainty and bias (as has been demonstrated in many simulations). Accounting for realistic data sparsity and likely spatial correlation requires new simulation frameworks that are not currently available for sablefish.	Development of a high-resolution OM for sablefish is a long-term research goal for the assessment team, but likely will require additional funding to complete. In the interim (coming year), the authors will explore further OM scenarios with the spatially explicit closed loop simulation model to identify potential benefits of increased aging across regions.
	Spatial Catch & Effort	Collect spatially explicit catch and effort data from logbooks for fixed gear fleet.	Would provide a spatially explicit fishery-dependent CPUE index and tracks spatial shifts in fishing grounds.	The logbook data is no longer being processed and would require additional resources to do so.	If logbook data becomes available, this recommendation will be revisited.
Spatial Research	Observer Data Analysis	Examine observer age compositions spatially.	Analysis of aging data collection locations will help determine if spatial shifts in fishing effort caused	Extensive analysis of otolith collection sites across years as well as vessels was undertaken prior to	Unfortunately, much of the analysis of AI observer data is confidential due to the limited

			historical "plus group" (age 31+) anomalies.	the CIE review in 2025. Interesting patterning was discovered, especially from the AI where periodic pulses of extremely large and old fish have been caught by the fishery. Based on observer data, it does appear that fishing effort shifts into the western AI drove the emergence of the large plus groups. However, it is unclear if subsequent decline in effort in that region was due to declining CPUE, declining value (gas prices and/or sablefish prices), or other socioeconomic factors.	number of active/observed vessels and the spatial nature of the data. Aggregated summaries of the data analysis will be developed and shared with review bodies in the near future. The authors plan to update the analysis annually as a way to track shifts in fishing effort distribution, locations of otolith collection spatially, and especially changes in dynamics in the AI.
	Stock Structure & Dispersal	Conduct larval dispersal modeling (MOM-6) across BS/AI and expand tagging (electronic/satellite).	Refines spatial movement parameters and connectivity assumptions for future spatial models.	Both of these recommendations are long-term research priorities for the assessment team but require additional funding to obtain the expertise to develop larval dispersal models and/or Hidden Markov Models (HMM) to analyze satellite tag tracks from geomagnetic fields.	The assessment team will continue to lead efforts and write grants to obtain funding for these recommendations.
	Added Operating Model Complexity	Include an additional population/region in the AI (i.e., a western AI region) within the spatially explicit operating model (OM), which only interacts with the fishing fleet in certain years.	Adding a region to the AI in the spatial simulation framework would help determine what influence a de facto spawning refugia (i.e., a western AI region composed of large, older fish that is only lightly fished in certain years) might have on assessment performance, especially in relation to the appearance of sporadic large plus groups in the fishery. It is noted that this would be a significant piece of work.	The authors have been in discussions on how to formulate such an OM for over a year but are primarily limited by issues with model conditioning for the new region given limited observed data. Primarily, initializing the population in this region as well as the level of movement out of this region are critical assumptions.	The authors are planning multiple research products based on the spatial closed loop simulations, one of which will include an additional western AI region. The results of the work will be presented when completed (~1-2 years).
	Make Spatial Model Operational	The Alaska-wide assessment should move towards a spatial model when it is sufficiently stable to provide reliable projections for calculating the reference points.	Given the extensive spatial complexity in sablefish biological and fishery dynamics, the spatial model is conceptually more appropriate to explicitly handle these spatial processes. Moreover, the considerable variation in catch across east, middle and west GOA (highest in the east),	The assessment team agrees that the spatial model has many appealing aspects and can better represent sablefish spatial dynamics. Extensive effort has been put into this model and continues, mainly to address issues related to spatially varying	The assessment team plans to provide annual updates to the five-region assessment model and will continue to explore the potential for its use as the basis of operational management advice, if and when it is deemed stable and robust (and when issues related to spatial

			would best be represented by a 5-area model rather than a 3-area model.	selectivity, time-varying movement, integration of electronic tagging data, and data weighting (tag data vs. other data inputs).	reference points and HCRs have been resolved).
	Spatial Reference Points	Develop methods to define and implement Tier 3 spatial management reference points and harvest control rules (HCRs).	Resolves uncertainty in how spatial reference points should be calculated and the scale at which they should be implemented (e.g., regionally vs. globally).	The assessment team is currently undergoing work to better define and understand the implications of different definitions of reference points in a spatial context. Because fish can now move among regions in a spatial model, defining equilibrium reference points becomes more complex. Additionally, many of the underlying issues must be determined by managers (e.g., should the HCR be applied globally or regionally) to guide how the authors should implement management catch projections.	The authors will continue simulation work exploring spatial reference points and HCRs and will update the review bodies (PT, SSC) on this work, as requested.
Software	Package Documentation	Write a formal technical design and architecture document for SPoRC.	Facilitates open collaboration, long-term maintainability, and developer succession.	This is an excellent recommendation to accompany the extensive documentation and vignettes currently available for SPoRC.	The lead developer will work on developing a formal design document for SPoRC in the coming year.
	Quality Assurance	Expand unit testing suite and automate routine (e.g., weekly) execution.	Prevents bug introduction and ensures long-term software stability as features are added.	Although already utilized for many code aspects, expansion of unit tests is a good suggestion.	The lead developer has already implemented this recommendation.

Table 8. Comparison of regional apportionment for 2023 based on the previous approach (i.e., extrapolating based on GOA for non-surveyed regions and including western AI extrapolated stations) and the updated approach that is consistent with the revised single region index construction (i.e., using carryover values for non-surveyed regions and removing the extrapolated western AI stations) used for Model *26.1_1Reg*. 2023 is used as a basis of comparison to ensure full consistency in approaches (given survey footprint change in 2025 and given that there was no survey in 2024) and are based on a five-year average including years 2019—2023. Note that apportionment values are rounded to the nearest hundredth in the table, but percent change is calculated on the raw values. Values for 2025 apportionment under the revised approach are also provided demonstrating minor changes due to updated survey data for 2025. Note that the WY and SEO apportionments do not include the 5% trawl allocation to SEO reallocated to WY (as is done in the final harvest specifications).

Region	2023 Apportionment Previous Approach (GOA Extrapolation)	2023 Apportionment Revised Approach (Carryover Values)	Percent Change from Old to New Apportionment for 2023	2025 Apportionment Revised Approach (Carryover Values)
BS	0.28	0.28	-1%	0.28
AI	0.24	0.19	-21%	0.20
WGOA	0.10	0.11	10%	0.11
CGOA	0.20	0.23	15%	0.23
WY	0.06	0.06	10%	0.06
SEO	0.12	0.13	11%	0.12

Figures

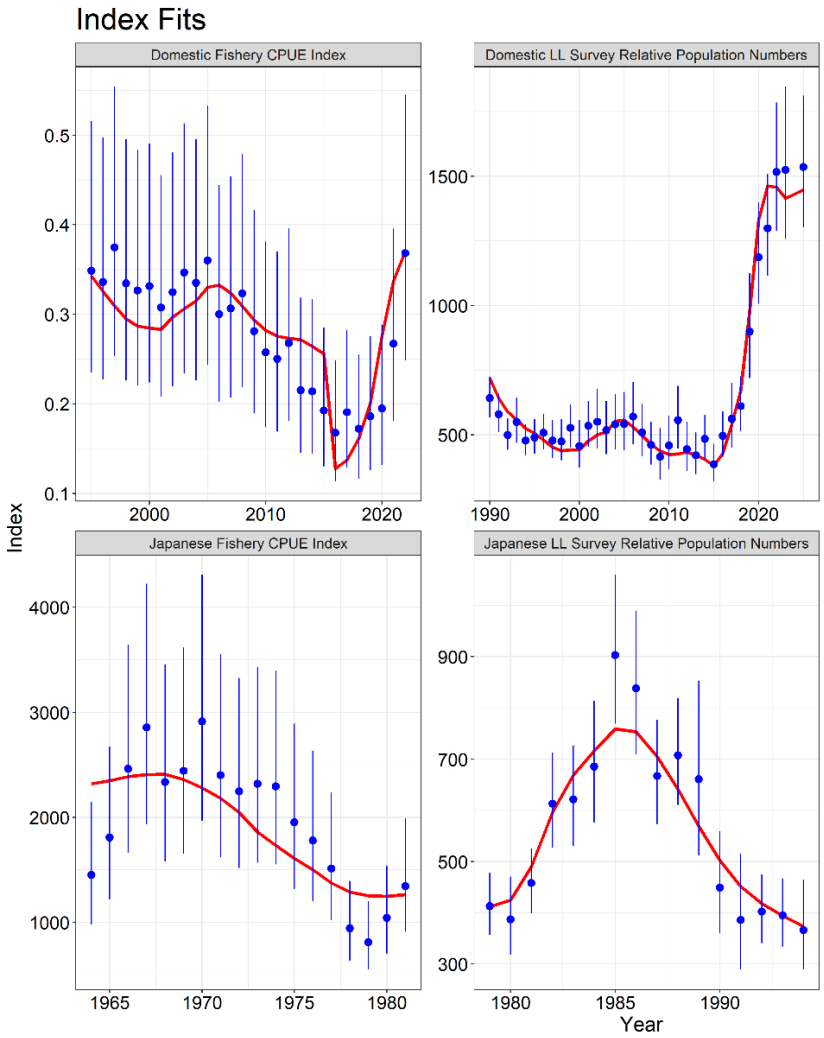


Figure 1. Model *25.12_1Reg_Cont* fit to indices, including the domestic fishery CPUE (top left), Japanese fishery CPUE (bottom left), NOAA Longline Survey RPNs (top right), and Japanese Longline Survey RPNs (bottom right). Red lines are model predicted values and blue dots are observed values (with 95% confidence intervals based on the assumed standard errors).

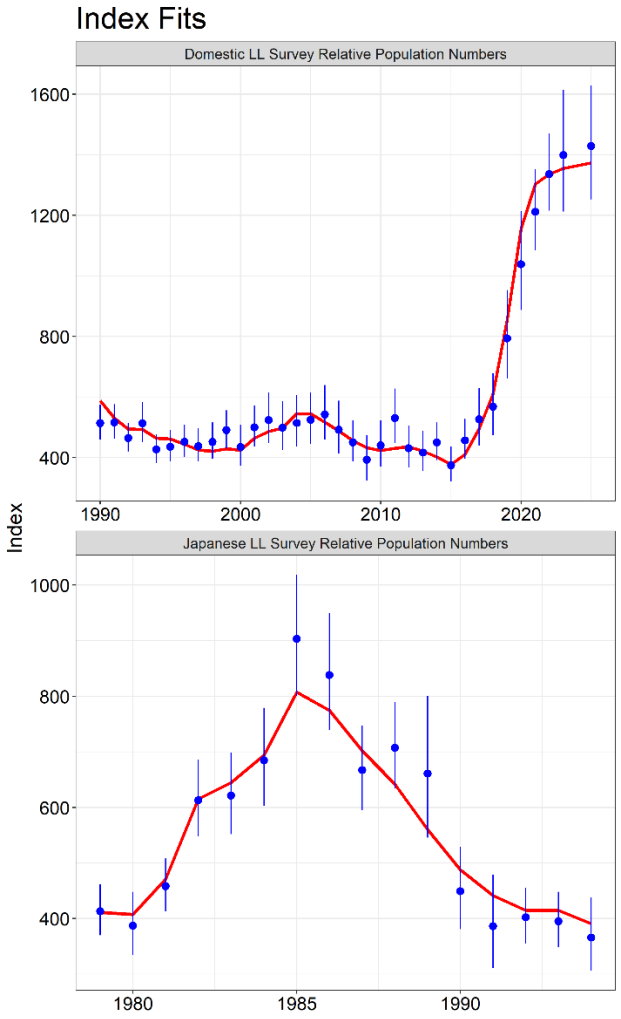


Figure 2. Model *26.1_1Reg* fit to indices, including the NOAA Longline Survey RPNs (top, 1000s of fish) and Japanese Longline Survey RPNs (bottom, 1000s of fish). Red lines are model predicted values, and blue dots are observed values (with 95% confidence intervals based on the assumed standard errors).

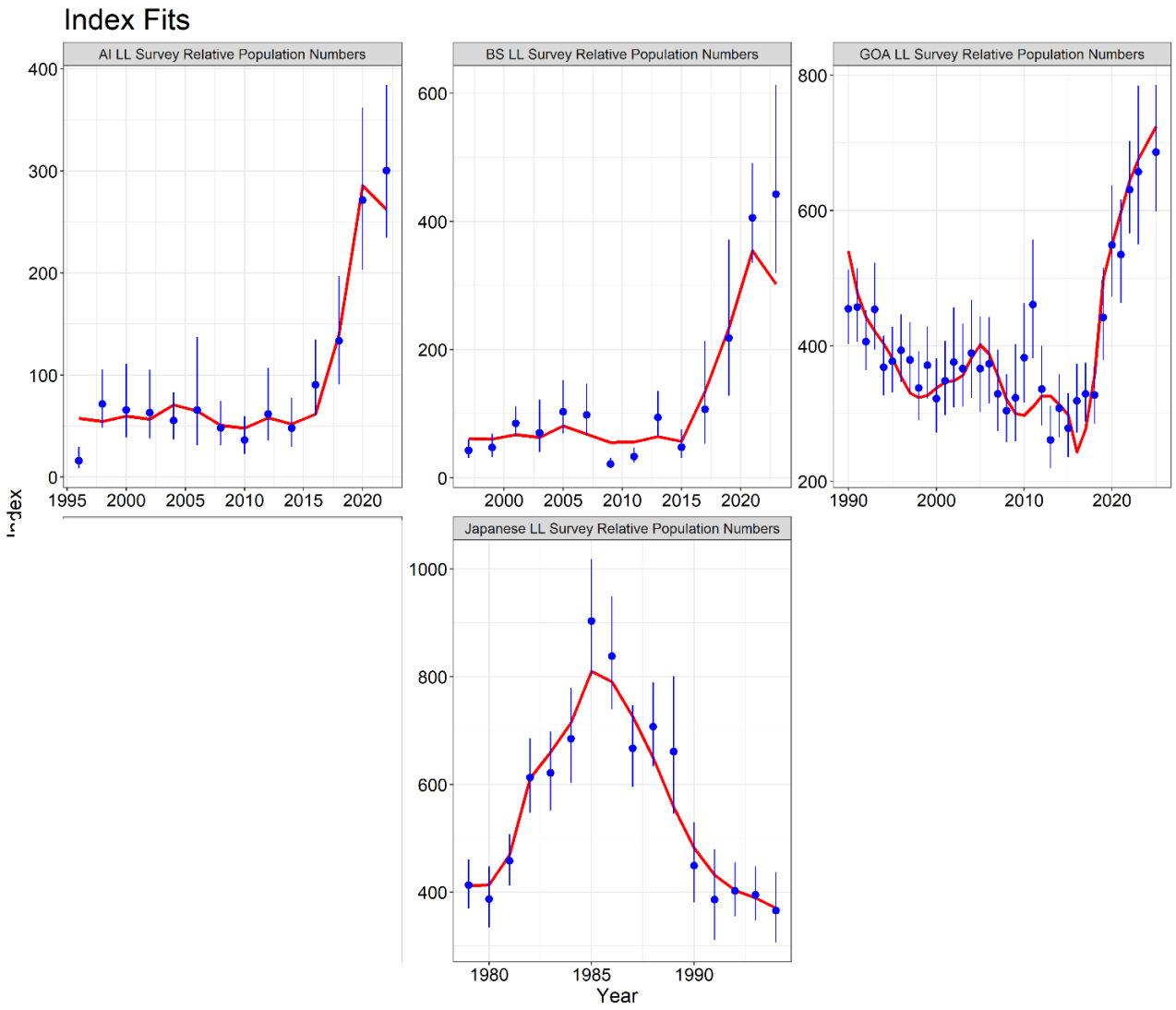


Figure 3. Model *26.2_FAA* fit to indices, including the NOAA Longline Survey RPNs (top, 1000s of fish) by region (columns) and Japanese Longline Survey RPNs (bottom, 1000s of fish). Red lines are model predicted values, and blue dots are observed values (with 95% confidence intervals based on the assumed standard errors).

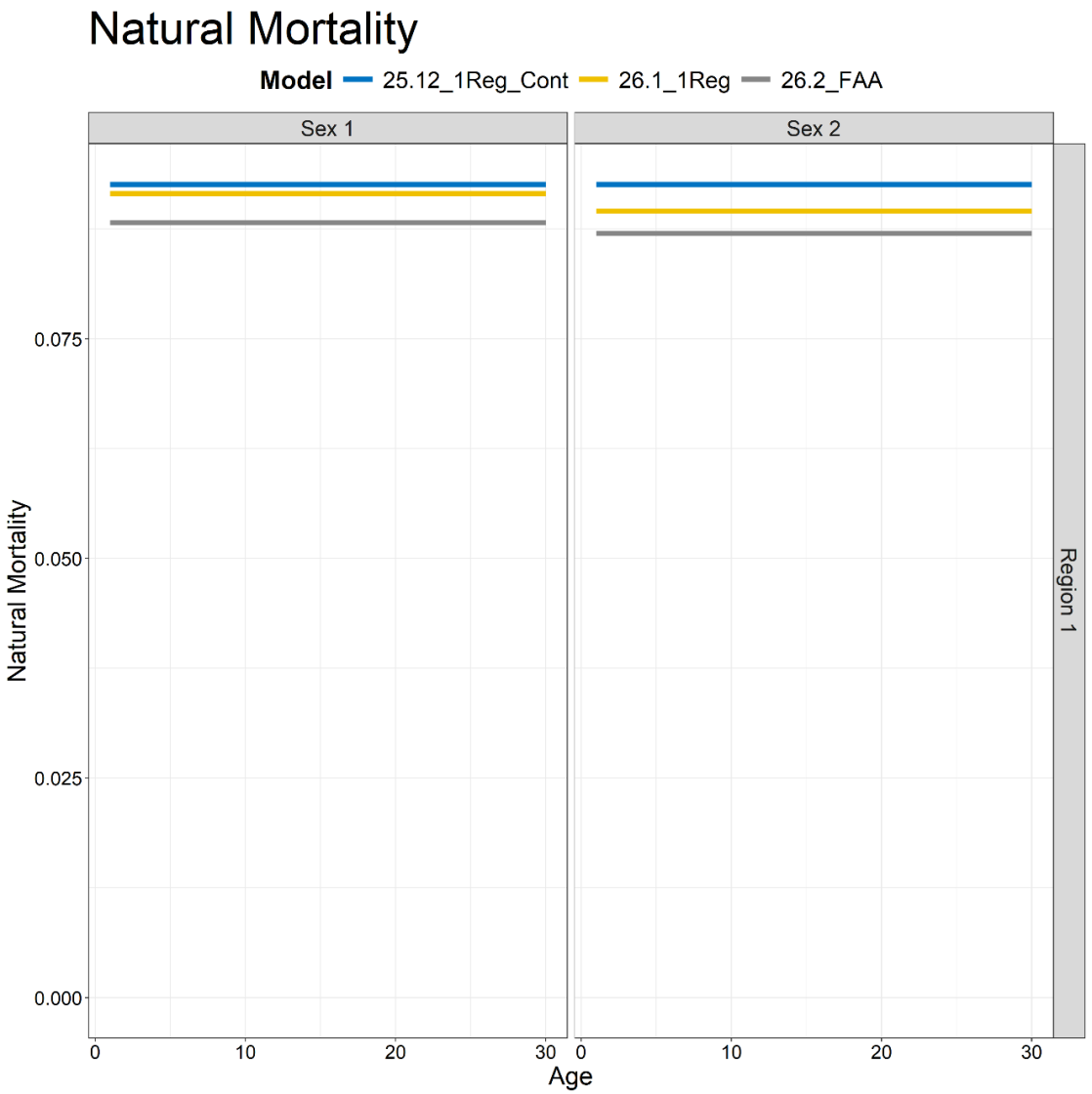


Figure 4. Natural mortality estimates from Model *25.12_1Reg_Cont* (blue line), Model *26.1_1Reg* (yellow line), and Model *26.2_FAA* (grey line). Sex 1 is females and sex 2 is males.

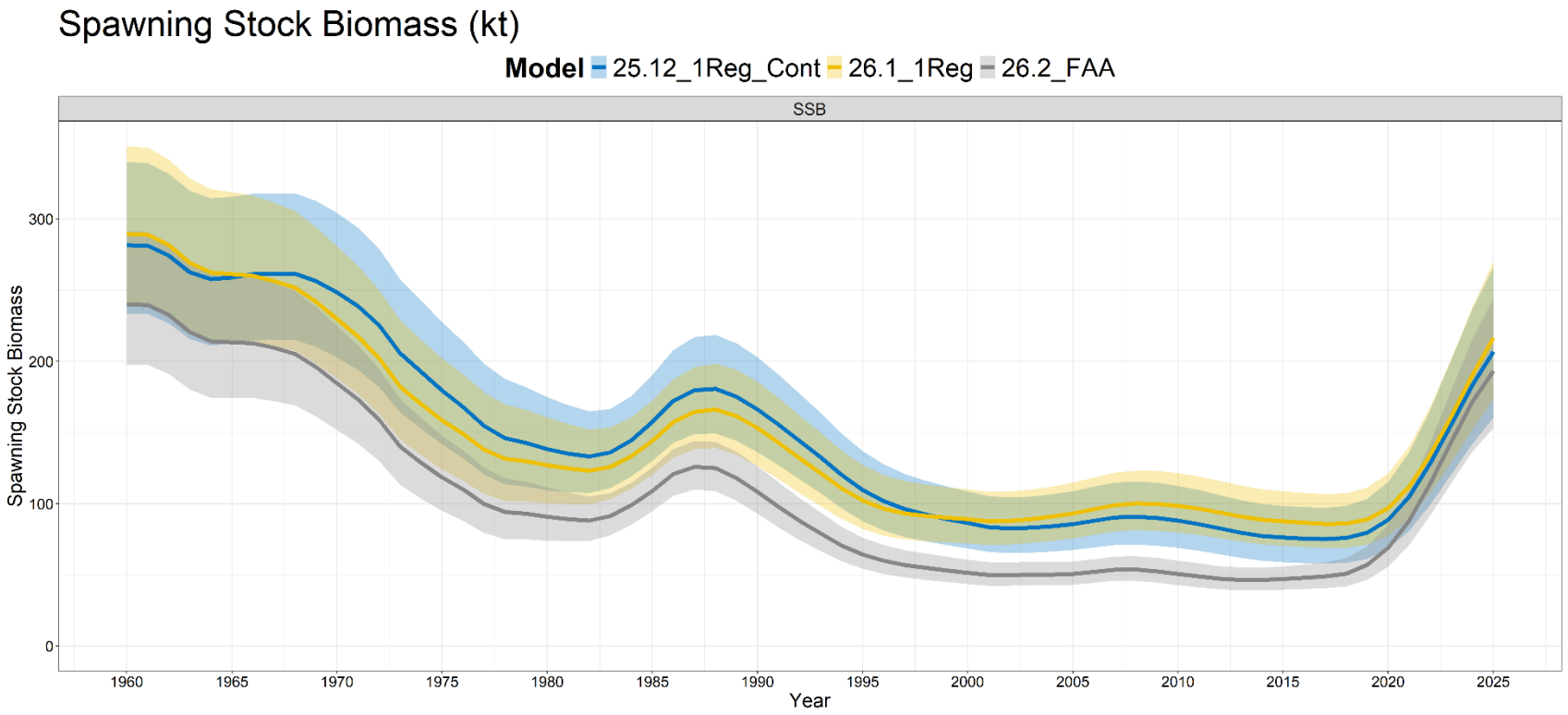


Figure 5. Comparison of spawning stock biomass (SSB, in kilotons) from Model *25.12_1Reg_Cont* (blue line), Model *26.1_1Reg* (yellow line), and Model *26.2_FAA* (grey line), where shaded regions represent the 95% confidence intervals.

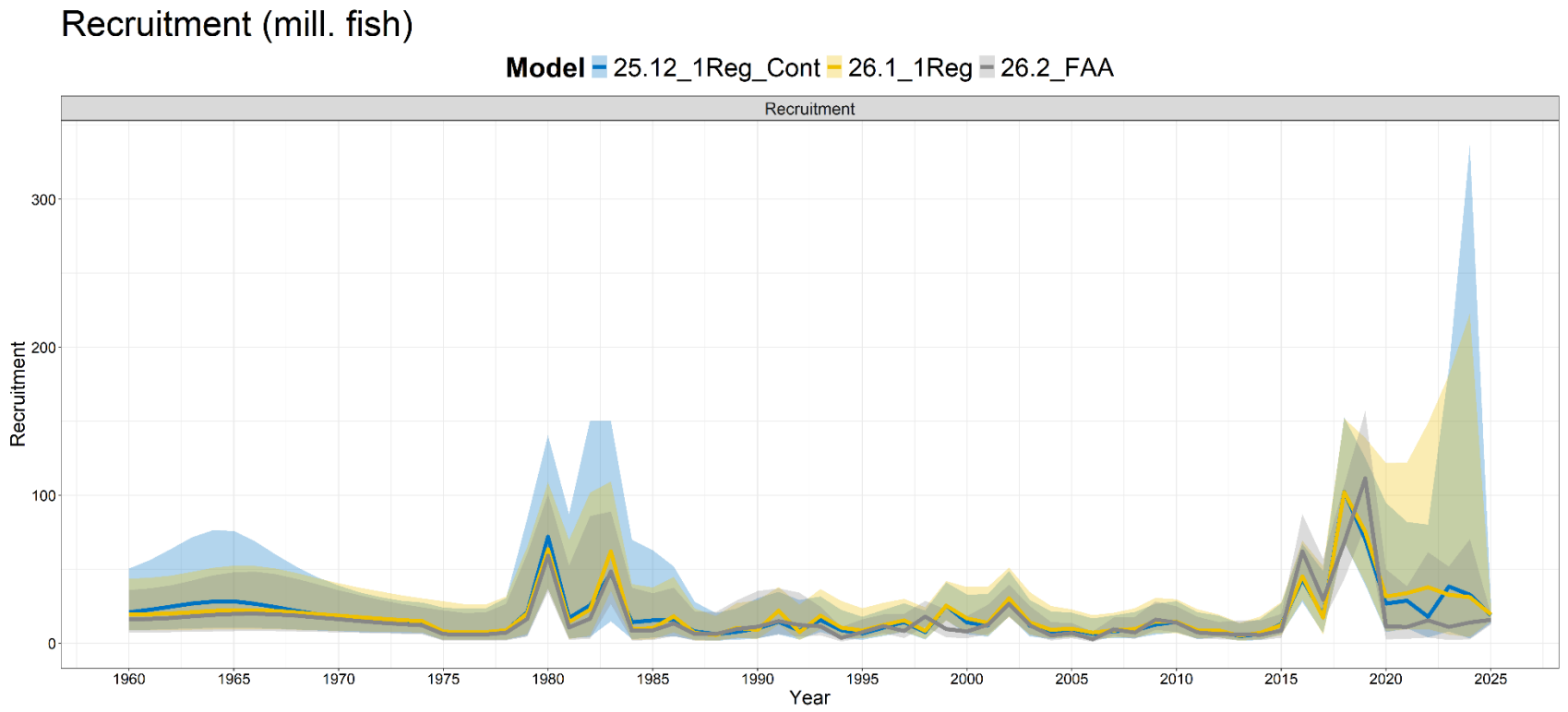
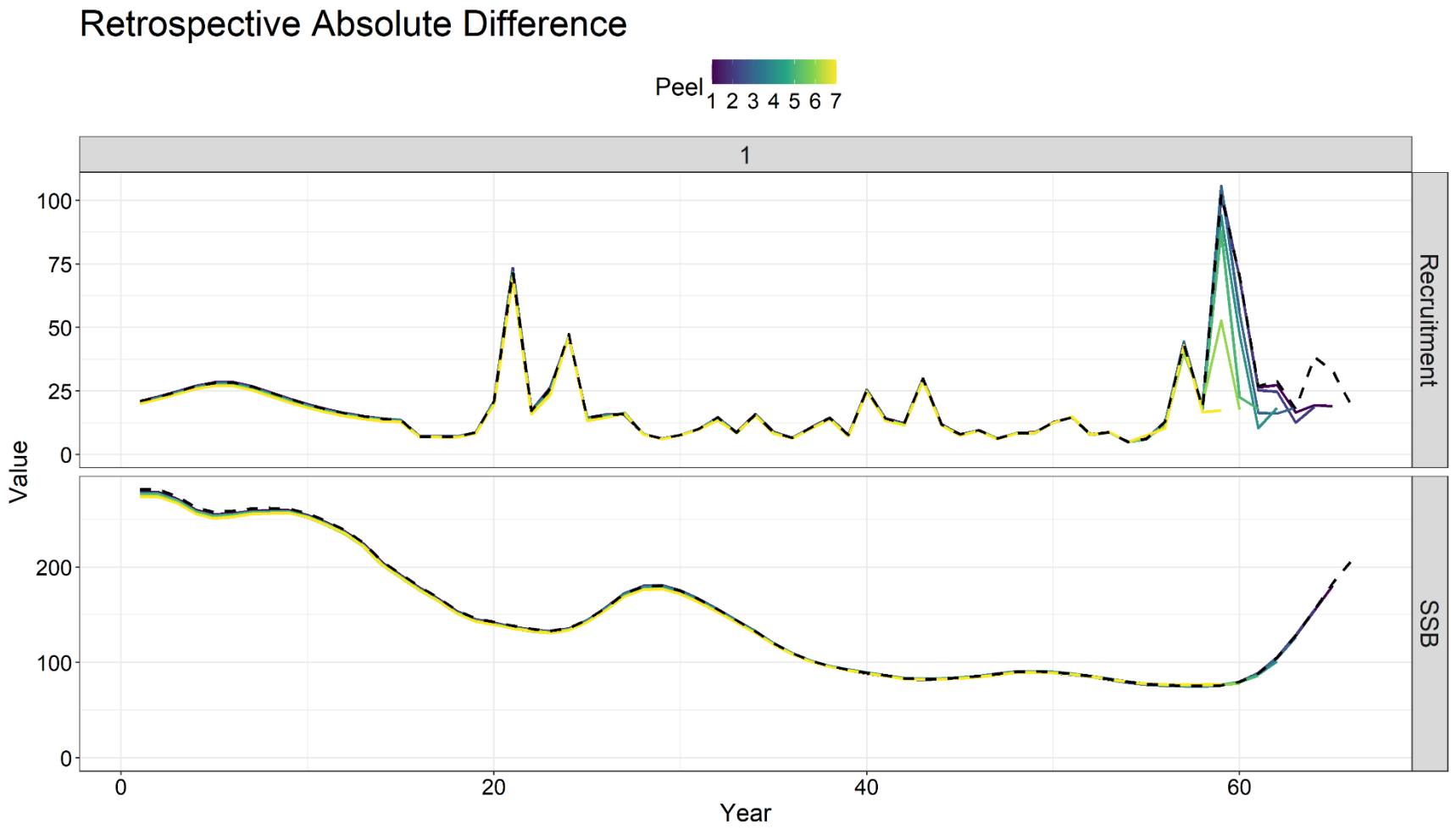


Figure 6. Comparison of recruitment (in millions of fish) from Model *25.12_1Reg_Cont* (blue line), Model *26.1_1Reg* (yellow line), and Model *26.2_FAA* (grey line), where shaded regions represent the 95% confidence intervals.



.

Figure 7. Comparison of recruitment (millions of fish; top panel) and spawning stock biomass (SSB; bottom panel) time series estimates from a seven year retrospective analysis for Model *25.12_1Reg_Cont*. Francis reweighting was undertaken for each peel to ensure consistency and to match how an assessment would be performed in a real-world situation. Mohn's rho for SSB was -0.014.

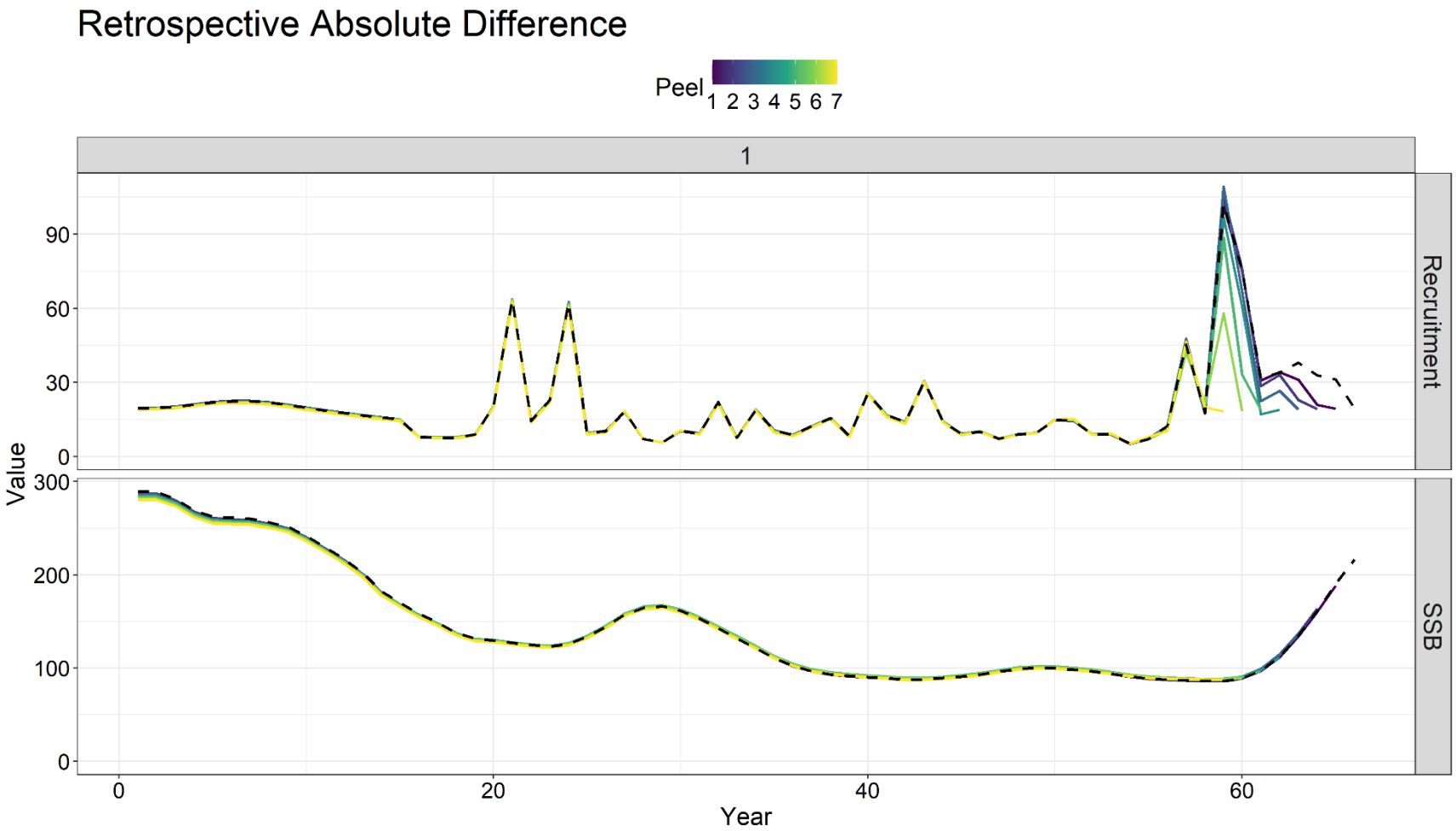


Figure 8. Comparison of recruitment (millions of fish; top panel) and spawning stock biomass (SSB; bottom panel) time series estimates from a seven year retrospective analysis for Model *26.1_1Reg*. Francis reweighting was undertaken for each peel to ensure consistency and to match how an assessment would be performed in a real-world situation. Mohn's rho for SSB was 0.0054.

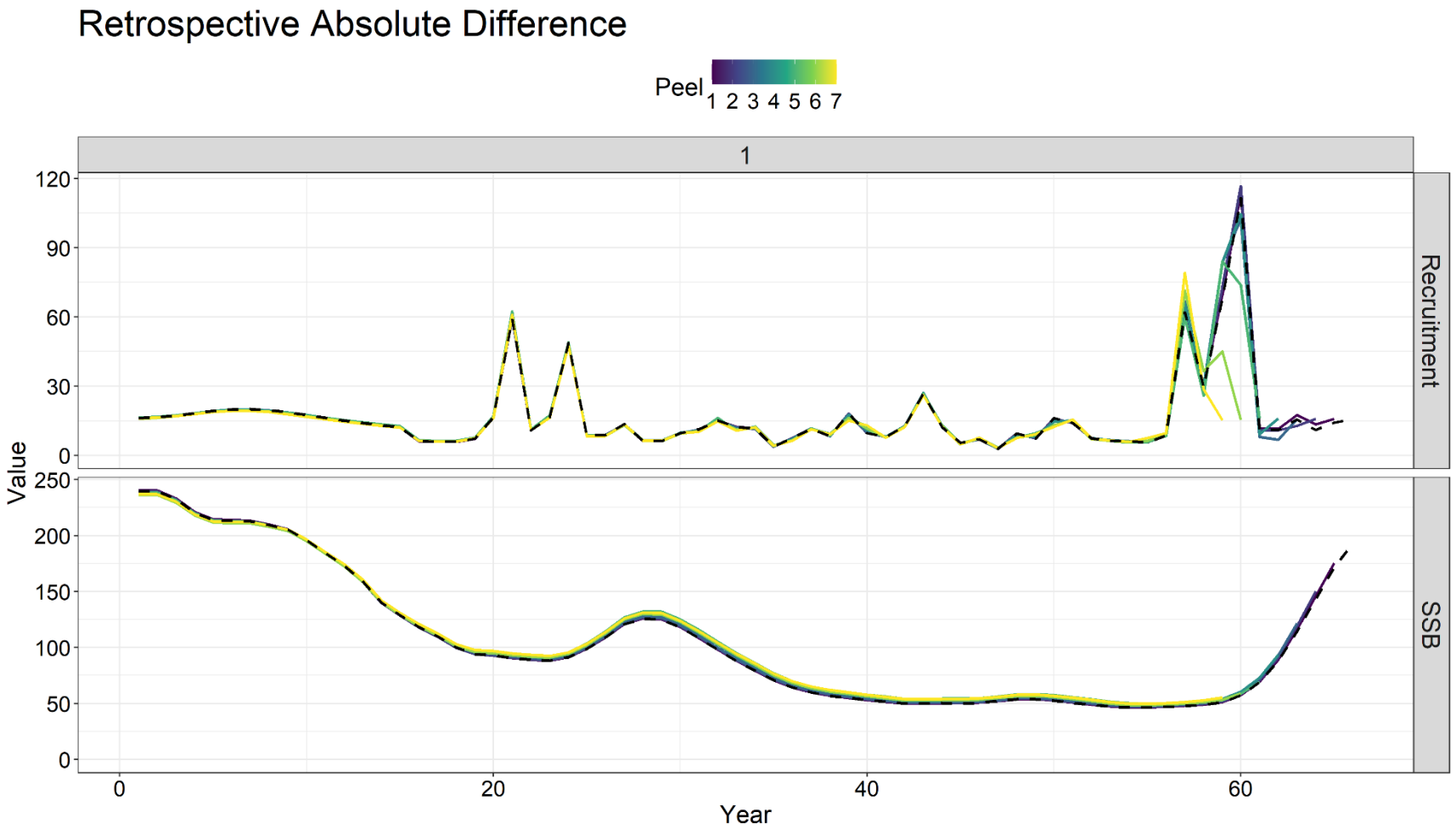


Figure 9. Comparison of recruitment (millions of fish; top panel) and spawning stock biomass (SSB; bottom panel) time series estimates from a seven-year retrospective analysis for Model *26.2_FAA*. Francis reweighting was undertaken for each peel to ensure consistency and to match how an assessment would be performed in a real-world situation. Mohn's rho for SSB was 0.046.

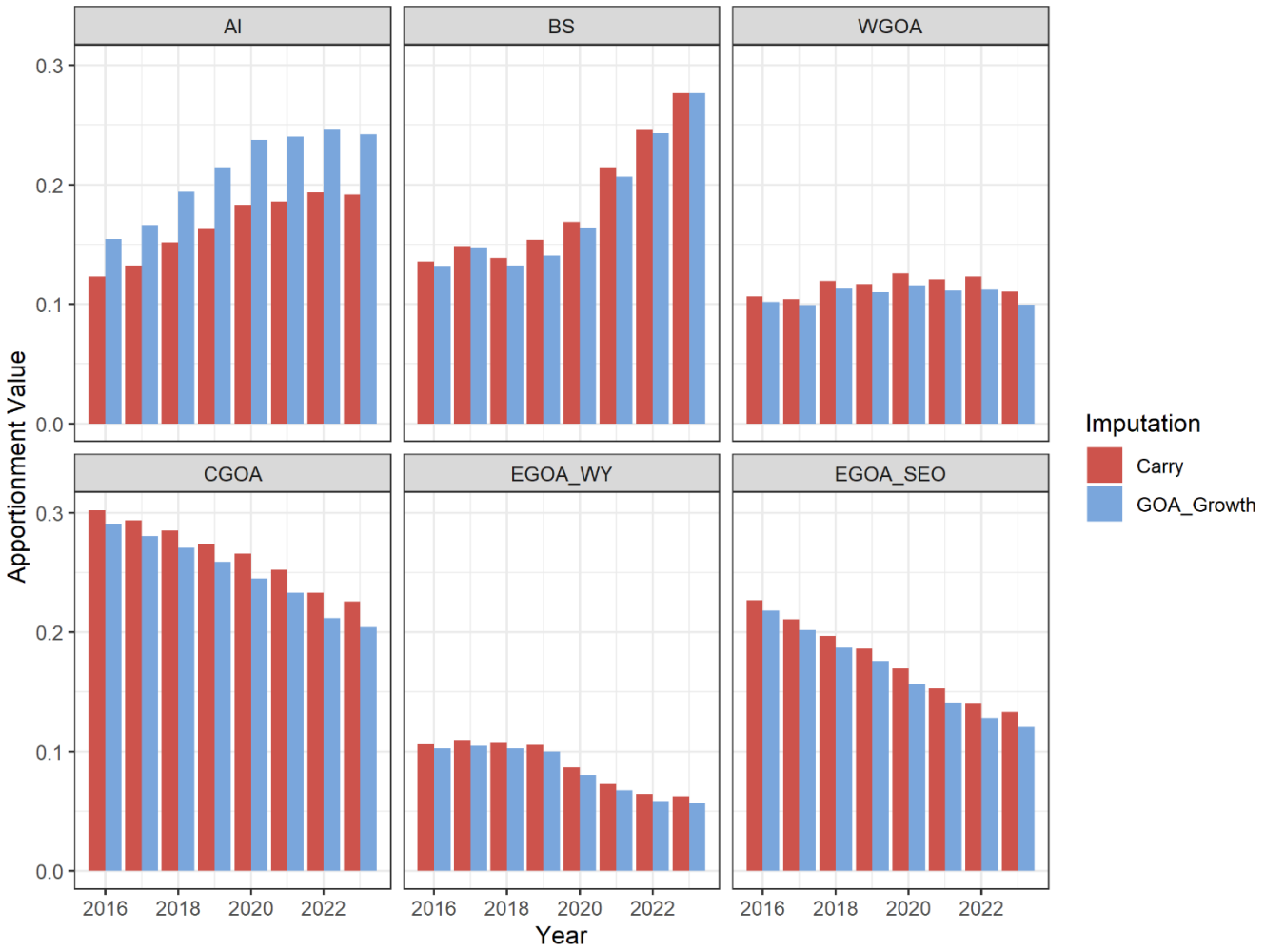


Figure 10. Comparison of regional apportionment from 2016 to 2023 for the previous approach (i.e., extrapolating based on the GOA year over year growth for non-surveyed regions and including western AI extrapolated stations; *GOA_Growth*) and the updated approach (i.e., using carryover values for non-surveyed regions and removing the extrapolated western AI stations; *Carry*). Note that the WY and SEO apportionments do not include the 5% trawl allocation to SEO reallocated to WY (as is done in the final harvest specifications).